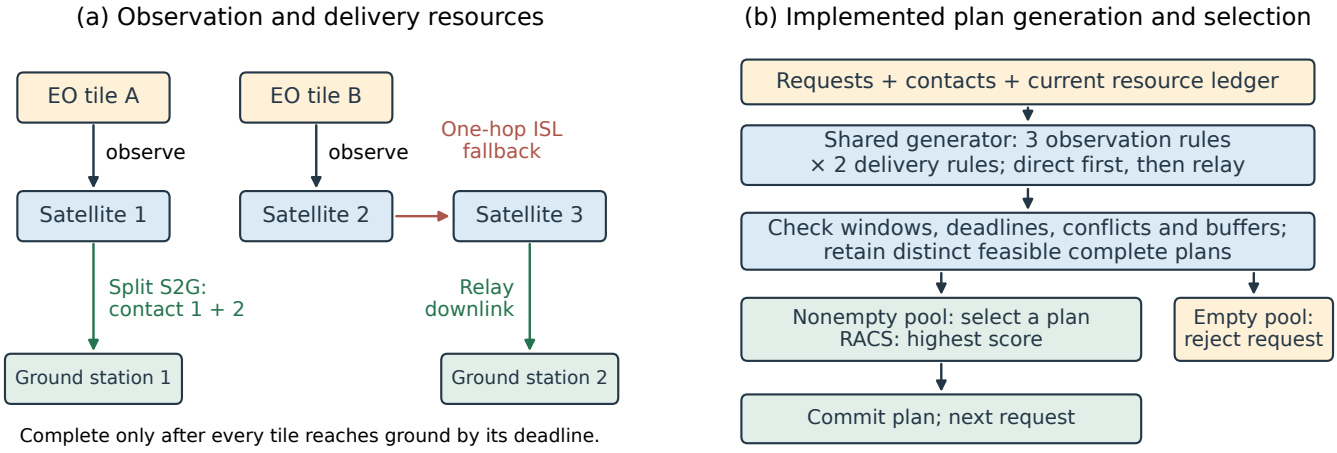
(a) Observation and delivery resources
EO tile A
EO tile B
observe
observe
One-hop ISL
fallback
Satellite 1
Satellite 2
Satellite 3
Split S2G:
contact 1 + 2
Relay
downlink
Ground station 1
Ground station 2
Complete only after every tile reaches ground by its deadline.
(b) Implemented plan generation and selection
Requests + contacts + current resource ledger
Shared generator: 3 observation rules
× 2 delivery rules; direct first, then relay
Check windows, deadlines, conflicts and buffers;
retain distinct feasible complete plans
Nonempty pool: select a plan
RACS: highest score
Empty pool:
reject request
Commit plan; next request